Sequence processing

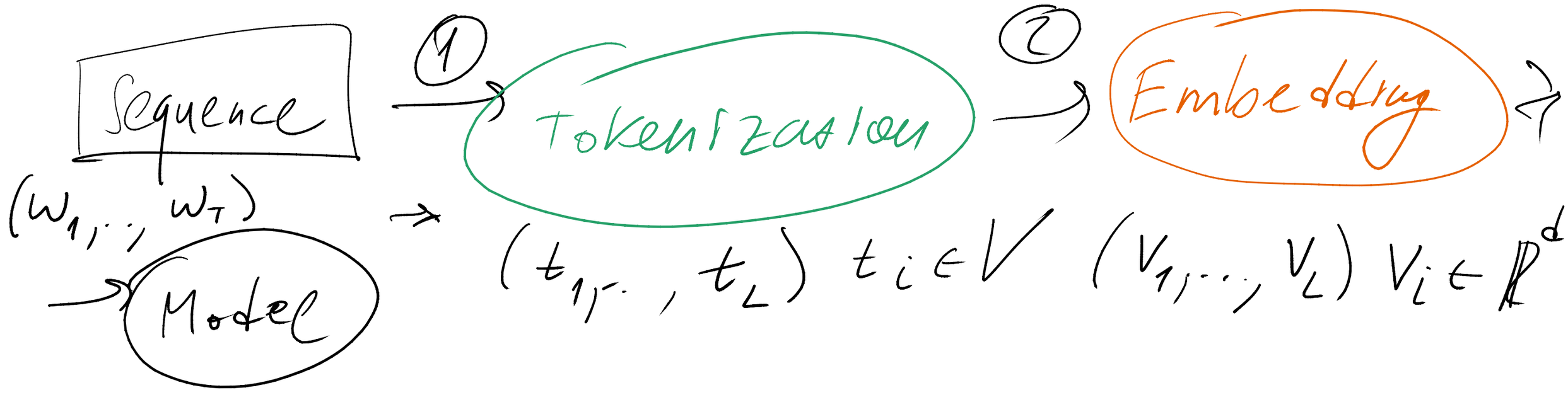
Processing pipeline

① Text is discrete

— μ x W x C float32

— ASCII Unicode

② Text variable length



1. Vocabulary <UNK>

2. Emb.

На дворе трава, на траве дрова.

- Character - level

| На | дворе | трава, | на | траве | дрова. |

+ very small vocab

+ easy to encode new words
− loses language semantics
− very long sequences

* Word-level | На | дворе | трава, | ...
− very large vocab
→ no tokens for new words

<LINK>

+ Preserve sentence structure
+ Shorter sequences

BPE (Byte-pair encoding)

Bottom up
ch. l → BPE → word-lev.

Unigram

ch. l. ← Unl ← word-lev

Top down

— BPE 1) a, b, c ... z, •, ␣ |V| small

2) While |V| < vocab_size:

- Find most frequent pair of tokens
- Merge 2 tokens to one, add to vocab

На дворе трава, на траве дрова.

0) {и, а, ж, в, о, р, е, т, ', ;, :, з}

1) иа, ав, тр, тра, ▽рав

- Unigram ALBERT

|V| large

p(sequence)

p(▽в|ЗА ▽аЗ х) << 1

p(▽прив▽т) > 0

$t_1, \ldots, t_L$ $p(t^{(i)}) = p^{(i)} = \frac{f_i}{f_1 + \ldots + f_{|V|}} \frac{z}{10}$

$p(t_1 \ldots, t_L) = p(t_1) \cdot \ldots \cdot p(t_L)$

$t^{(1)}, t^{(2)} \ldots t^{(|V|)}$ → max

↑ choose $\not{t}^{(i)}$ with max $p(t_1 \ldots t_L)$ without it

$s_1, s_2 \dots s_N$ $\quad s_i = (t_{11} \dots, t_{21})$

Corpus

$$\sum_{i=1}^{N} \sum_{j=1}^{L_i} -\log p(t_j | i) \to \min$$

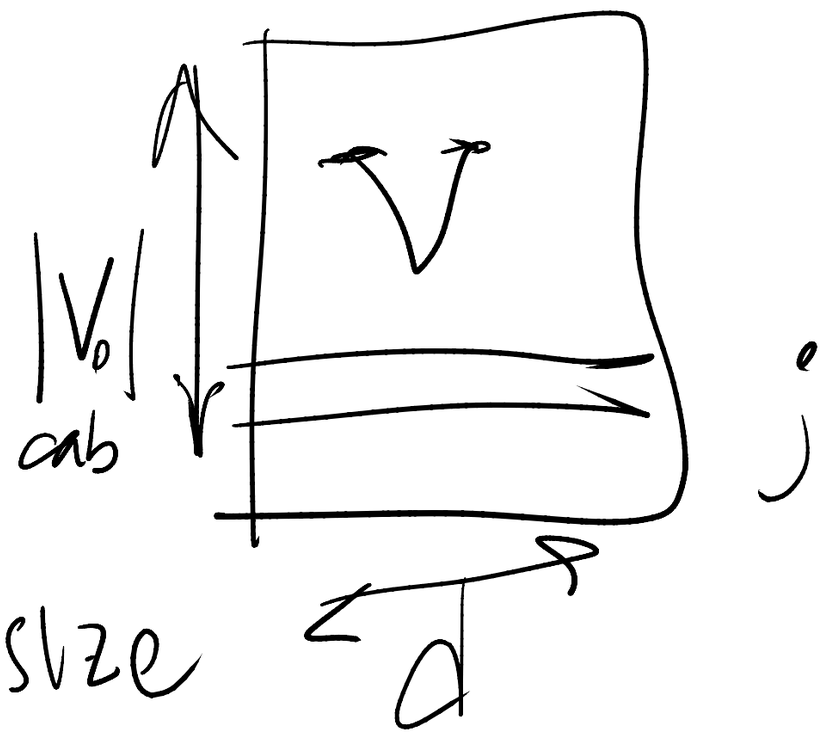
Word2Vec (history)

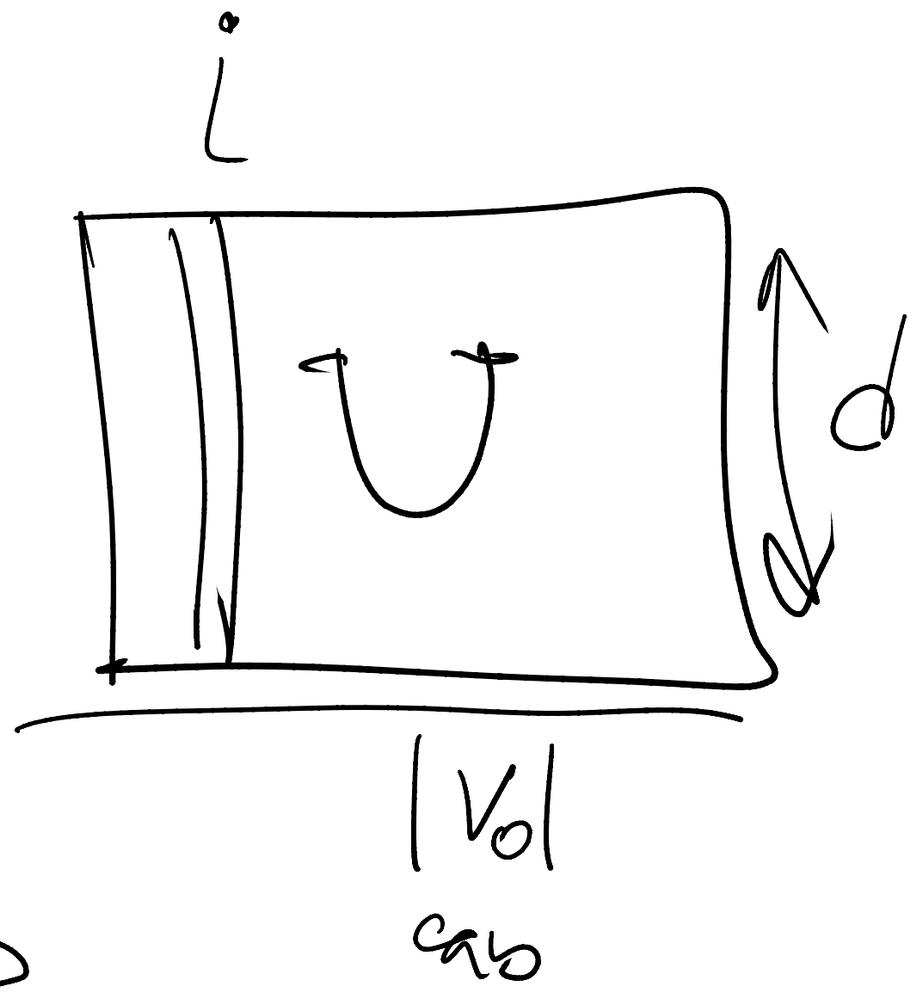
Politician Senator

$w_1 [w_2\ w_3\ w_4\ w_5\ w_6]\ w_7$

- Continious bag of words (CBOW)
- Skip-gram $\uparrow P(w_4 | w_2, \dots w_6)$

$$p(w_2 \dots, w_6 | w_4) \approx p(w_2 | w_4) \cdot \dots \cdot p(w_6 | w_4)$$





$$\langle v_j, u_i \rangle$$

$$p(w_i | w_j) \simeq \frac{\exp(v_j^T u_i)}{\sum_{k=1}^{|V_{vocab}|} \exp(v_j^T u_i)}$$

$$-\sum_{w_j - \text{center}} \log p(w_i | w_j) \to \min_{V, U}$$

w_i - context

$$p(w_j | w_{j-H}, w_{j-H+1}, \dots, w_{j-1}, w_{j+1}, \dots, w_{j+H})$$

$$= (V_{j-H} + V_{j-H+1} + \dots + V_{j+H})^T u_j \quad \text{for CBOW}$$

Binary class

Negative sampling

$$-\sum \log\left(\sigma\left(v_j^T u_i\right)\right) - \sum \log\left(1-\sigma\left(v_j^T u_i\right)\right) \rightarrow \min_{V,U}$$

w_j - center

w_i - context

w_i, w_j — random words

$V, U^T, \frac{1}{2}(V + U^T)$

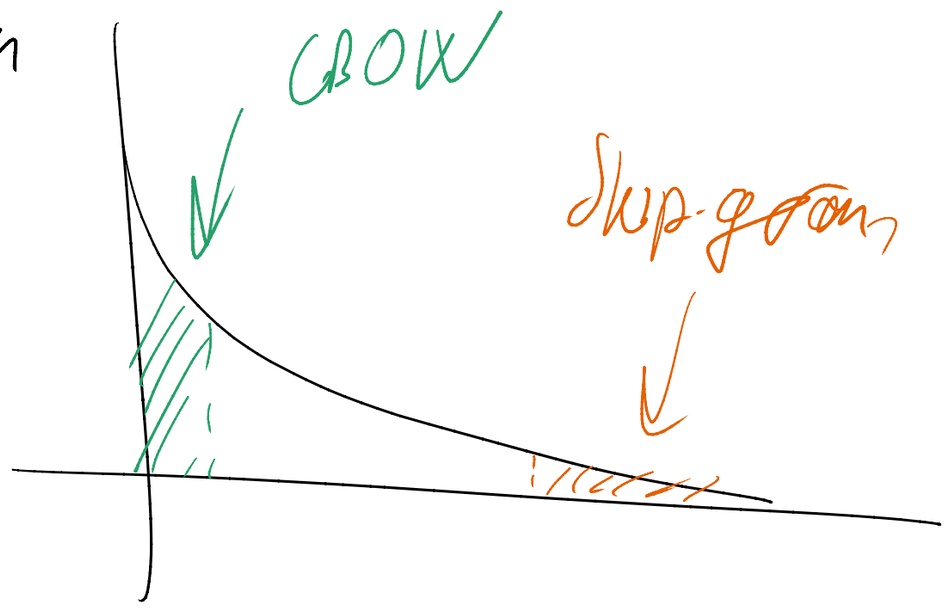
↑ ↑ ↑

$V_{man} - V_{king} \approx V_{woman} - V_{queen}$

neg sampling

CBOW frequent

Skip-gram rare



FastText (***)

W2V (Google)

p(добрый | добрый)

p(добрый | доб)

p(добрый | до)

| ...

| добр

| обр

| ...

$w_4 \mid w_5 \mid w_6$

$p(w_4 \mid w_5)$ . .

n-gram

$n = 3 \ldots 6$

+ sum of n-gram embeds — new embed

+ info exchange between words

Glove (Global vectors)